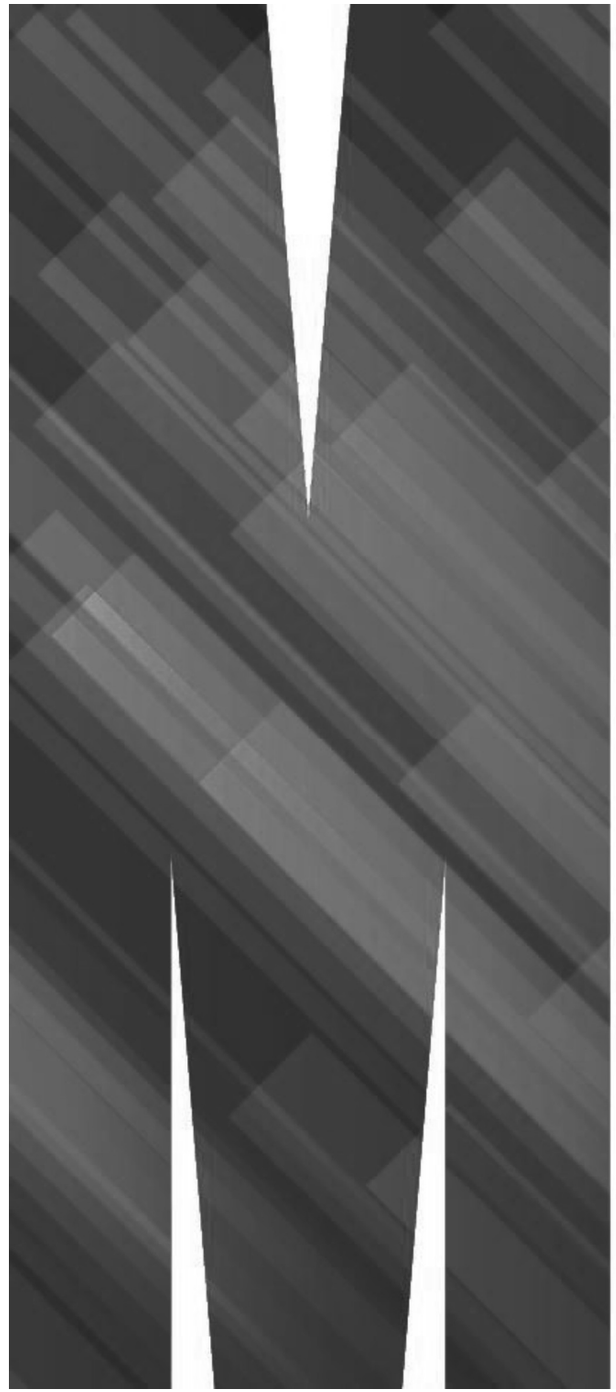
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Topic 6.2
Regression Models

Numerical Inputs transformation and Categorical inputs coding and consolidation

By Dr Lee How Chinh



Objectives

- Explain the effect of extreme and unusual values, and use transformation to handle these values.
- Handle categorical inputs for regression models.

Handling extreme or unusual values

Handling extreme or unusual values

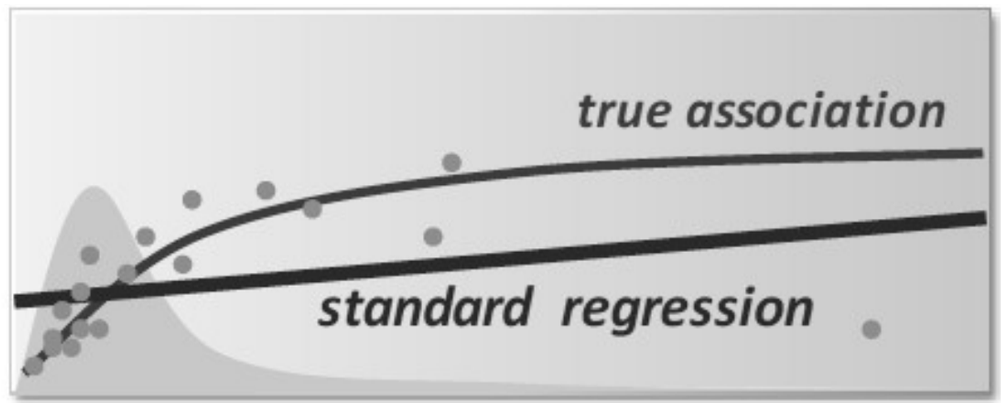
- Linear regression has no assumptions on the distribution of inputs.
- So why should we worry about the extreme input distributions?

Handling extreme or unusual values

There are at least two reasons.

First reason:

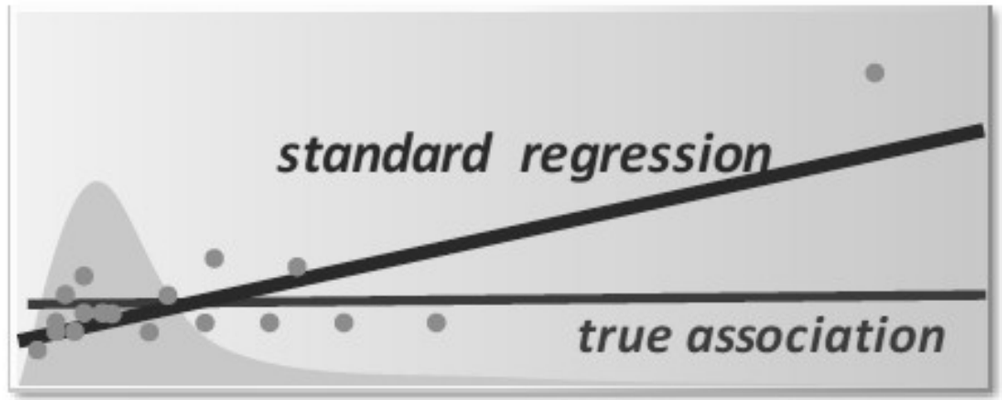
- In most real-world applications, the relationship between expected target value and input values does not increase without bound.
- It will tapers off to some horizontal asymptote.
- Standard regression models are unable to accommodate this relationship.



Handling extreme or unusual values

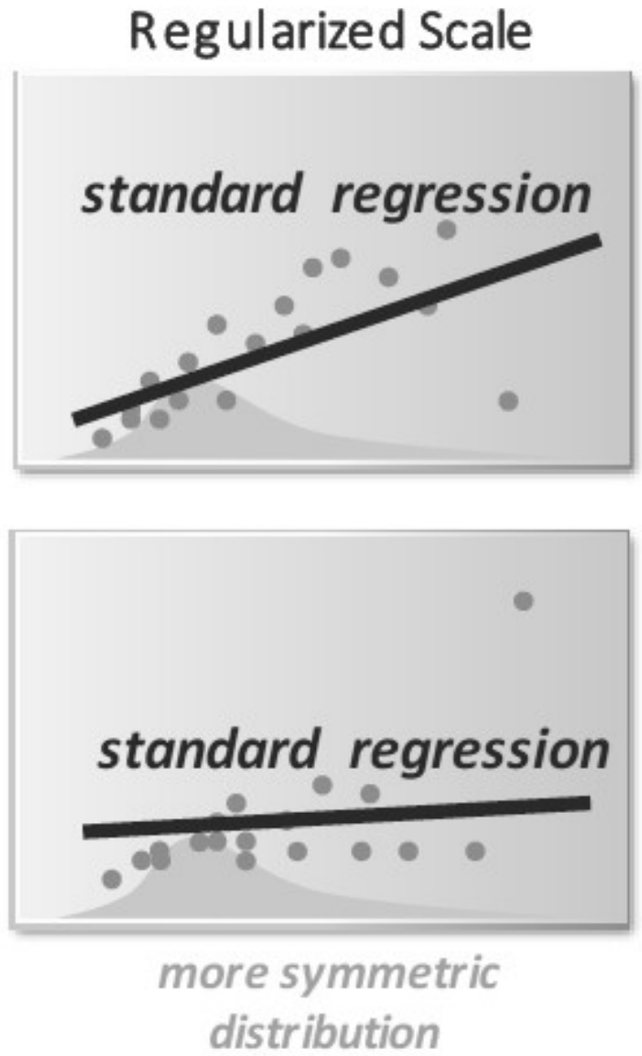
Second reason:

- A leverage point is a point that expands from the overall mean of a distribution.
- Models built on inputs with extreme distributions attempt to optimize fit the most extreme points.
- This can result in an exaggeration of an input's association with the target.

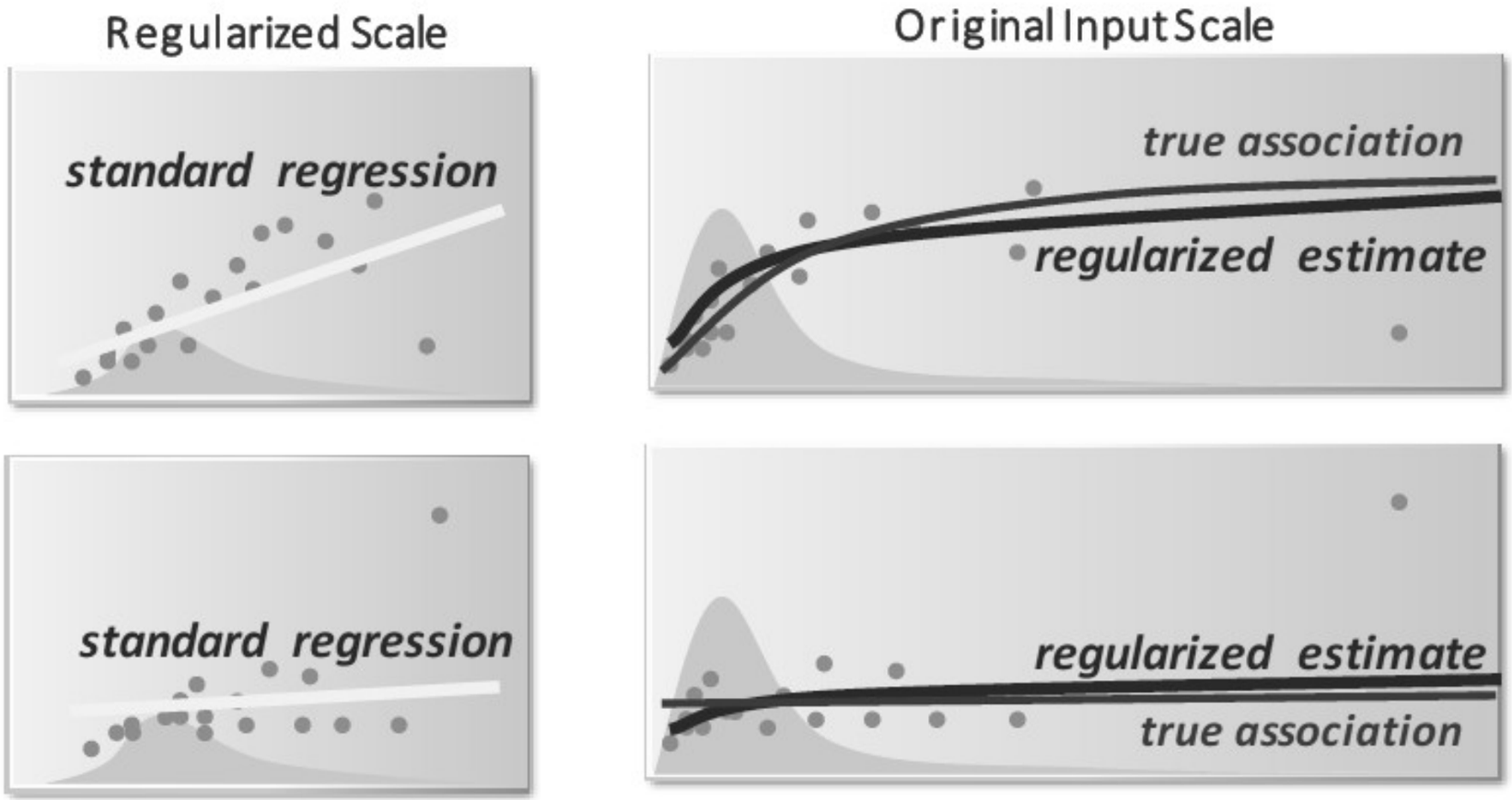


Handling extreme or unusual values

- A more straightforward and practical approach is transforming the inputs to eliminate extreme values.
- Then, a standard regression model can be fit using the transformed input in place of the original input.



Handling extreme or unusual values



Categorical Inputs

Categorical Inputs

- To include the categorical inputs in a regression model, we must convert the categorical inputs into some numeric values.
- The most common method is creating design variables or dummy variables, with each design variable representing approximately one level of the categorical input.

Coding Redundancy

Level	D_A	D_B	D_C	D_D	D_E	D_F	D_G	D_H	D_I
A	1	0	0	0	0	0	0	0	0
B	0	1	0	0	0	0	0	0	0
C	0	0	1	0	0	0	0	0	0
D	0	0	0	1	0	0	0	0	0
E	0	0	0	0	1	0	0	0	0
F	0	0	0	0	0	1	0	0	0
G	0	0	0	0	0	0	1	0	0
H	0	0	0	0	0	0	0	1	0
I	0	0	0	0	0	0	0	0	1

Level	D_A	D_B	D_C	D_D	D_E	D_F	D_G	D_H
A	1	0	0	0	0	0	0	0
B	0	1	0	0	0	0	0	0
C	0	0	1	0	0	0	0	0
D	0	0	0	1	0	0	0	0
E	0	0	0	0	1	0	0	0
F	0	0	0	0	0	1	0	0
G	0	0	0	0	0	0	1	0
H	0	0	0	0	0	0	0	1
I	0	0	0	0	0	0	0	0

Coding Consolidation

- A single categorical input can significantly increase a model's degrees of freedom and increase the chances of overfitting.
- One of the simplest way is to reduce the number of levels of the categorical input.

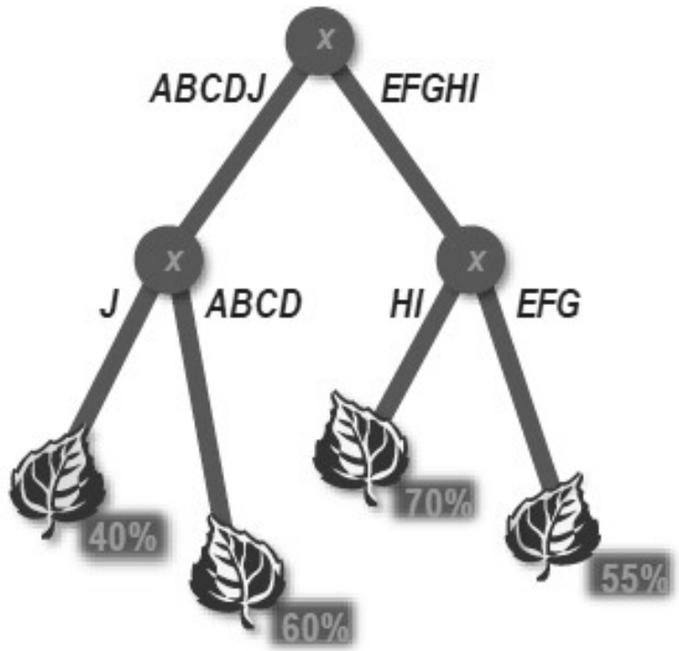
Coding Consolidation

Level	D_A	D_B	D_C	D_D	D_E	D_F	D_G	D_H
A	1	0	0	0	0	0	0	0
B	0	1	0	0	0	0	0	0
C	0	0	1	0	0	0	0	0
D	0	0	0	1	0	0	0	0
E	0	0	0	0	1	0	0	0
F	0	0	0	0	0	1	0	0
G	0	0	0	0	0	0	1	0
H	0	0	0	0	0	0	0	1
I	0	0	0	0	0	0	0	0

Level	D_{ABCD}	D_B	D_C	D_D	D_{EF}	D_F	D_{GH}
A	1	0	0	0	0	0	0
B	1	0	0	0	0	0	0
C	1	0	0	0	0	0	0
D	1	0	0	0	0	0	0
E	0	0	0	0	1	0	0
F	0	0	0	0	1	1	0
G	0	0	0	0	0	0	1
H	0	0	0	0	0	0	1
I	0	0	0	0	0	0	0

Using Decision Tree to Consolidate Categorical Input

- Decision trees can easily group the distinct levels of the categorical variable.
- Build a decision tree using the categorical variable of interest as the sole modelling input.
- The split search algorithm then groups input levels with primary outcome proportions.
- The IDs for each leaf replace the original levels of the input.



Lesson Summary

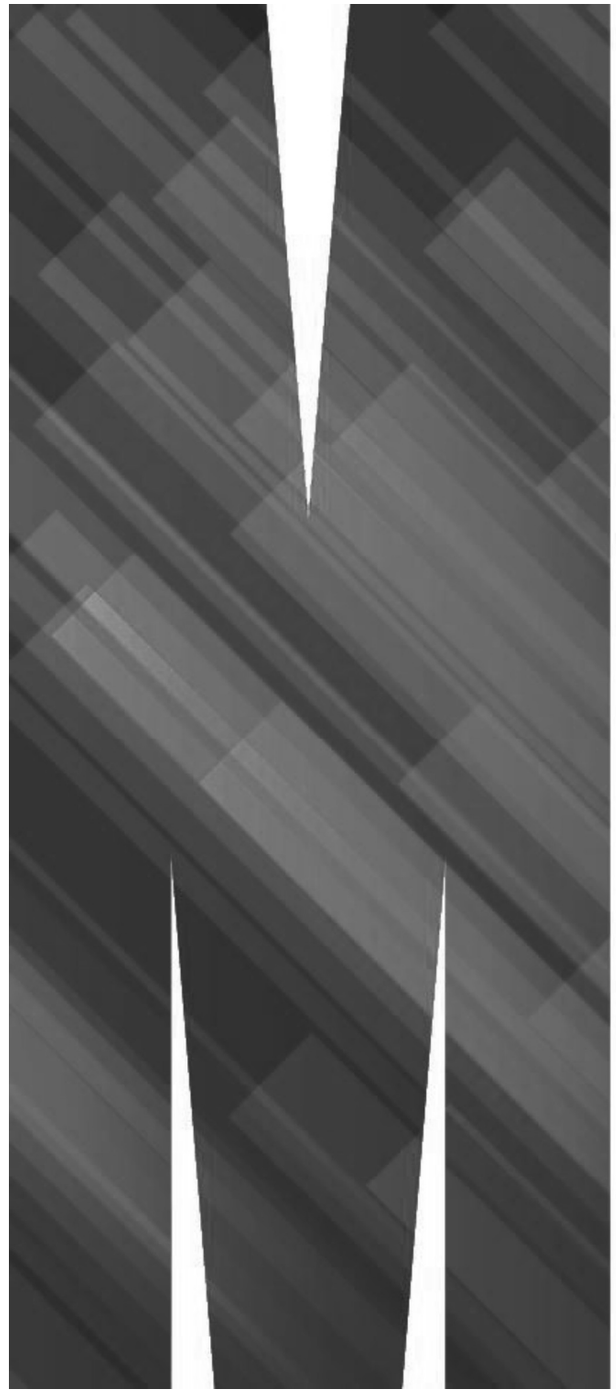
- Input Transformation is used to handle the relationship with horizontal asymptote and distribution with leverage points.
- Design variable or dummy variable is used to convert categorical inputs into numeric values.
- A categorical input with many levels can increase the chances of overfitting.
- A simple method to overcome this problem is to reduce the number of levels of the categorical input.
- A decision tree can group the levels of a category input.

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Topic 6.3
Regression Models

The three elements of Generalised Linear Models

By Dr Lee How Chinh



Objectives

- Identify the difference between General Linear Models and Generalised Linear Models
- Name the three elements of Generalised Linear Models.

Generalised Linear Models

General Linear Models Vs Generalised Linear Models

General Linear Models have a continuous response and agree to some assumptions.

There are four assumptions associated with general linear models.

- The relationship between the mean of response variable, Y, and predictor variables, Xs, is linear.
- The random errors are normally distributed
- The variance of the random errors is constant.
- Observations are independent.

General Linear Models Vs Generalised Linear Models

The General Linear Model is not suitable if,

- The relationship between predictor variables, Xs and response variable, Y is not linear.
- Variance of the random errors is not constant and varies with predictor variables.
- Response variable is discrete or categorical.

General Linear Models Vs Generalised Linear Models

The assumptions for Generalised Linear Models are,

- The observations are independent and random
- The response variable, Y, is from an exponential family
- The original response variable need not have a linear relationship with independent variables, but the transformed response variable is linearly related to independent variables.

Three Elements of Generalised Linear Models

Three Elements of Generalised Linear Models

- A generalised linear model is an extension of a general linear model.
- It enables the population mean to depend on a linear predictor through a nonlinear link function.

Three Elements of Generalised Linear Models

$$g(\mu_i) = \beta_0 + \sum_{i=1}^{k} \beta_i x_i$$

The three components of Generalised Linear Models

1. Systematic Component/ Linear Predictor
   - The linear combination of the predictors and the regression coefficients
2. Random Component
   - the probability distribution from an exponential family of the response variable.
3. Link Function, $g(\mu_i)$
   - specifies the function between the random and systematic components

Three Elements of Generalised Linear Models

- A generalized linear model requires a probability distribution from an exponential family and a link function.
- The probability distribution should match the distribution of the response variable.
- The link function is used to related the response variable to the predictor (effect) variables.

Three Elements of Generalised Linear Models

- The distribution of the response variable imposes range requirements on the response variable.

Distribution	Range Requirements
Beta	Values must be between *0* and *1*, exclusive.
Binary	Exactly two distinct values
Exponential	Nonnegative real values
Gamma	Nonnegative real values
Geometric	Positive integers
Inverse Gaussian	Positive real values
Negative Binomial	Nonnegative integers
Normal	Real values
Poisson	Nonnegative integers

Three Elements of Generalised Linear Models

- Some common link functions for the respective distributions

Link	$\eta_i = g(\mu_i)$
Identity	μ_i
Log	$\log_e \mu_i$
Inverse	μ_i^{-1}
Inverse-square	μ_i^{-2}
Square-root	$\sqrt{\mu_i}$
Logit	$\log_e \frac{\mu_i}{1-\mu_i}$
Probit	$\Phi^{-1}(\mu_i)$
Log-log	$-\log_e[-\log_e(\mu_i)]$
Complementary log-log	$\log_e[-\log_e(1-\mu_i)]$

Beta	Logit, Probit, Log-log, C-log-log
Binary	Logit, Probit, Log-log, C-log-log
Exponential	Log, Identity
Gamma	Log, Identity, Reciprocal
Geometric	Log, Identity
Inverse Gaussian	Power(-2), Log, Identity
Negative Binomial	Log, Identity
Normal (default)	Identity, Log
Poisson	Log, Identity

Lesson Summary

- General linear models follow four strict assumptions. General linear models fail to produce unbiased estimates when the data violates the assumptions.
- Generalised linear models are the extension of general linear models where the assumptions are more relaxed compared to general linear models.
- Generalised linear models depend on three components; systematic component, random component and link function
- The link function transforms the expectation of the response variable that has an exponential family probability distribution to the linear function of predictors variables.

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Which generalised linear model is the most popular in business analytics?



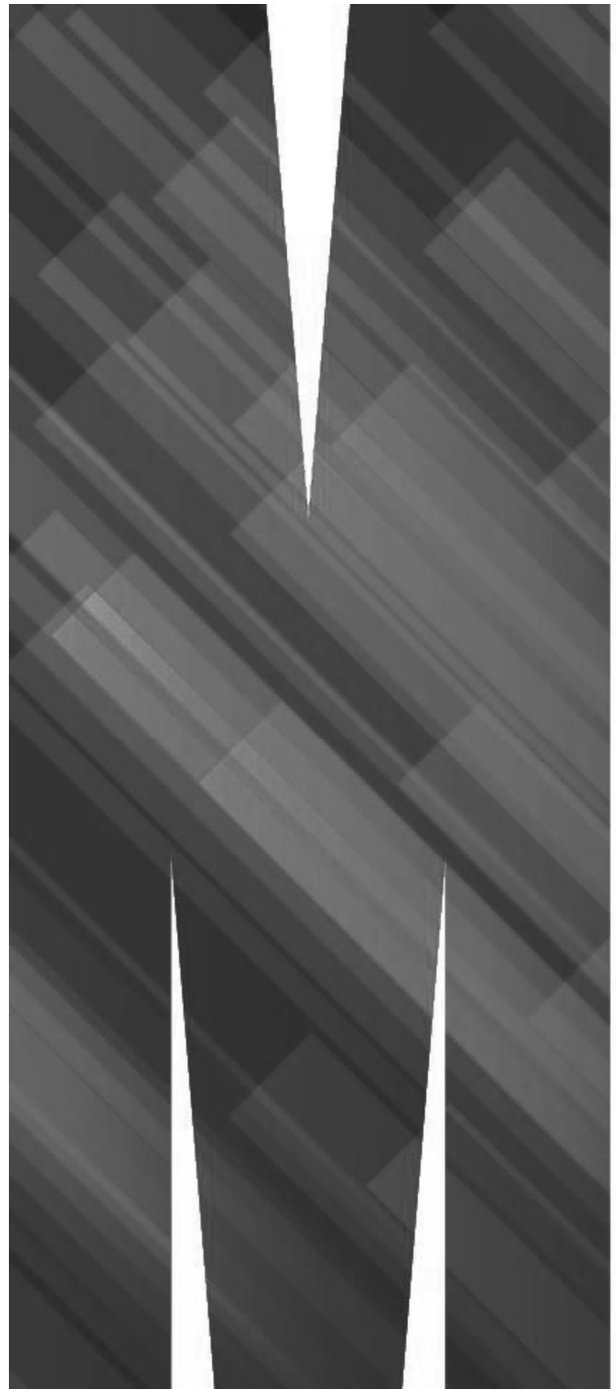
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Topic 6.4
Regression Models

Logistic regression and why it is important to business analytics

By Dr Lee How Chinh



Objectives

- Recognise the importance of logistic regression models in business analytics
- Describe and explain the three components of a logistic regression model .

Logistic regression models in business analytics

Logistic regression models in business analytics

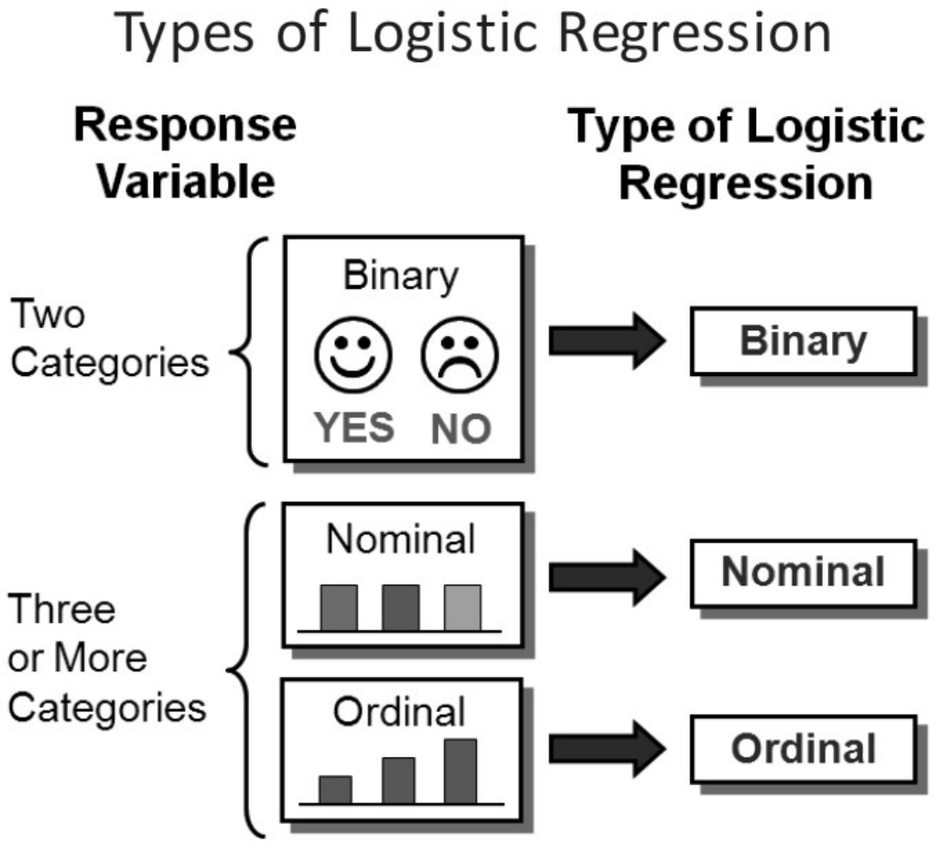
Logistic regression is a powerful modelling technique used extensively in business analytics due to the following reasons;

- The logistic regression allows categorical response and both numeric and categorical predictor variables.
- It can be used for both prediction and analytical analysis.
- It can produce decision, rank and estimate type predictions.
- It allows more complex models with interaction terms and higher-order terms.

Logistic regression models

Three Elements of Logistic Regression Models

- Logistic regression allows categorical response for more than two levels and ordinal response variable.
- In this chapter, we mainly focus on the binary response variable.

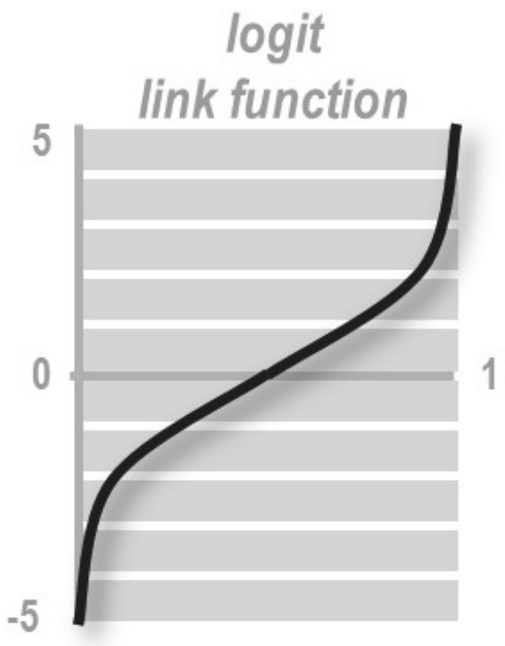


Three Elements of Logistic Regression Models

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

The response of logistic regression has a binomial distribution and a logit link function.

Three Elements of Logistic Regression Models



$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

The logit link function transforms probabilities (between 0 and 1) to logit scores (between $-\infty$ and $+\infty$).

Three Elements of Logistic Regression Models

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \text{logit}(\hat{p}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

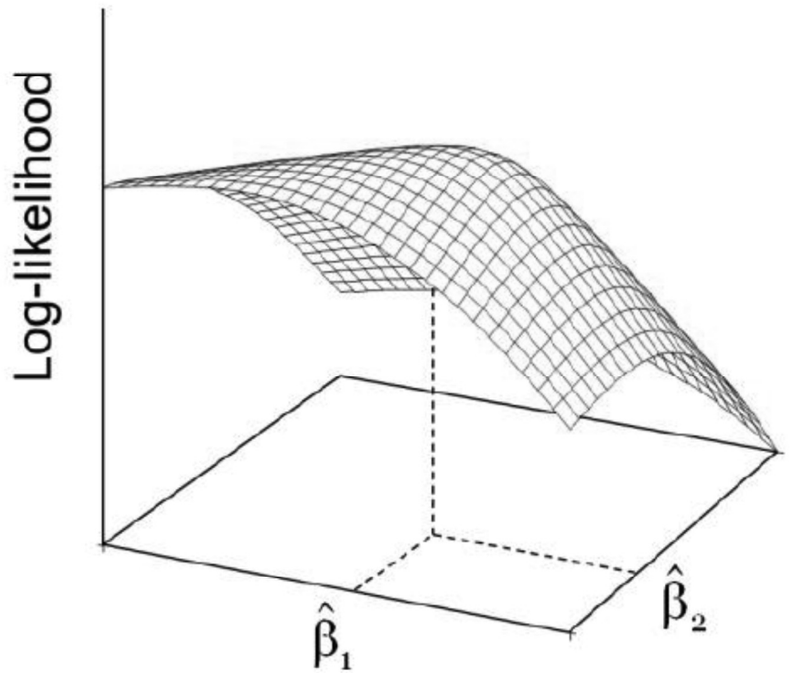
$$\hat{p} = \frac{1}{1+e^{-\text{logit}(\hat{p})}} = \frac{1}{1+e^{-(\hat{\beta}_0+\hat{\beta}_1 x_1+\cdots+\hat{\beta}_k x_k)}}$$

To obtain prediction estimates, the logit equation is solved for $\hat{p}$.

Three Elements of Logistic Regression Models

$$\text{logit}(\hat{p}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

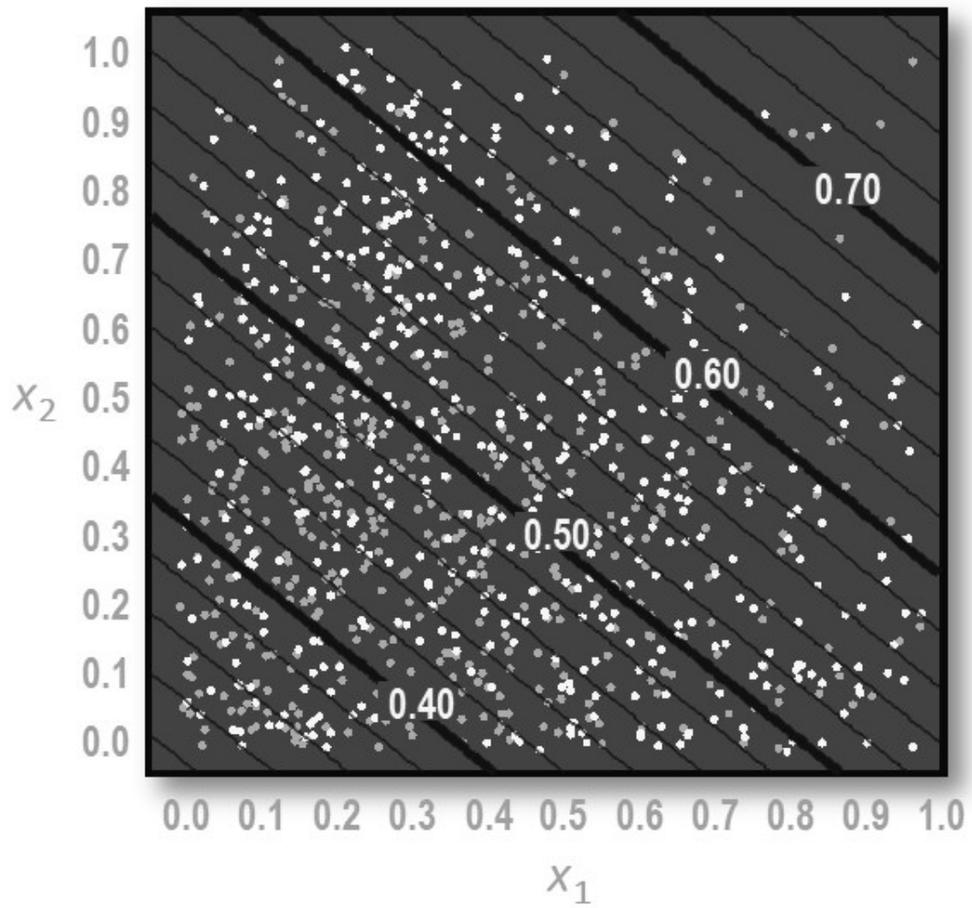
- For prediction, we need intercepts and parameter estimates.
- The parameter estimates are obtained from maximizing the log-likelihood function.



Logistic regression prediction illustration

$$\text{logit}(\hat{p}) = -0.81 + 0.92x_1 + 1.11x_k$$

- The prediction formula is obtained from maximum likelihood estimation.



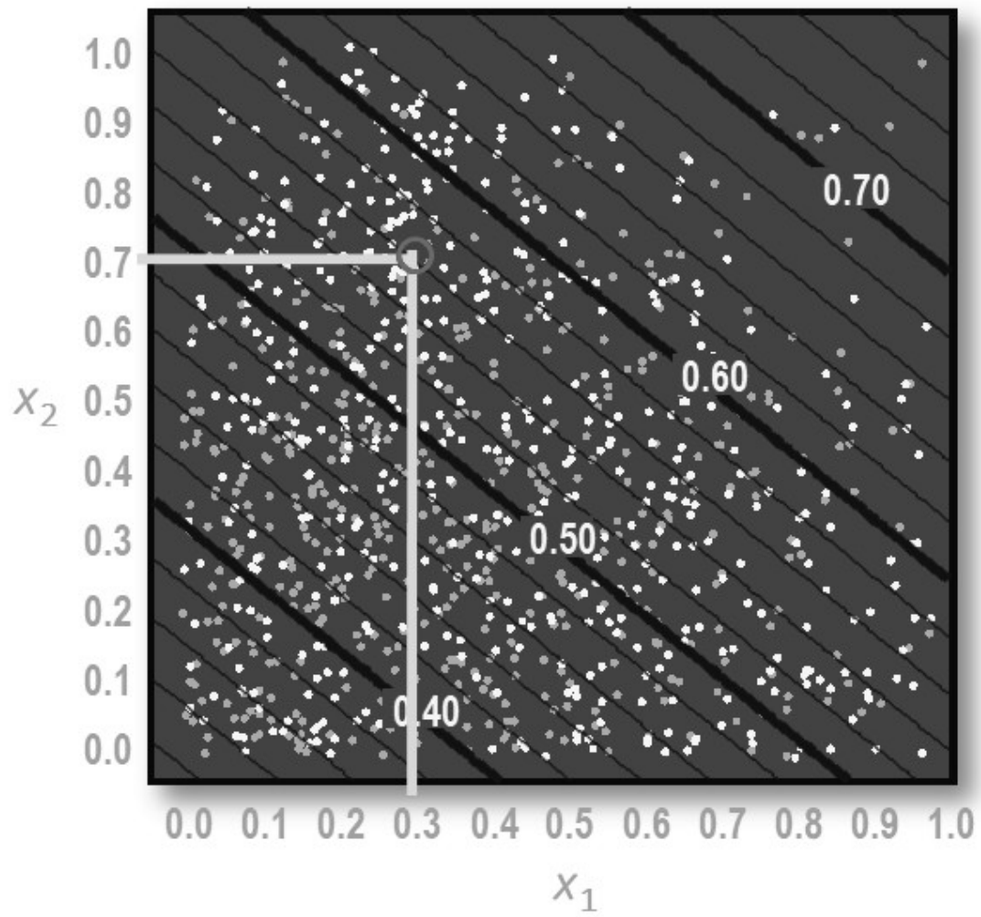
Logistic regression prediction illustration

$$\text{logit}(\hat{p}) = -0.81 + 0.92x_1 + 1.11x_k$$

$$\hat{p} = \frac{1}{1 + e^{-\text{logit}(\hat{p})}}$$

What is the logistic regression prediction when $x_1 = 0.3$ and $x_2 = 0.7$?

a) 0.243
b) 0.56
c) Yellow
d) It depends



Lesson Summary

- Logistic regression is useful in business analytics because it has a categorical response variable, can be used for decision, rank and estimate prediction, and is easily interpreted. Logistic regression can be extended to a more complex model with interaction and higher-order terms.
- A logit link function is used to transform a binomial distributed response variable.
- The maximum likelihood estimation is used to find the parameter estimates.

How can I interpret the prediction and explain the relationship between response and predictors?



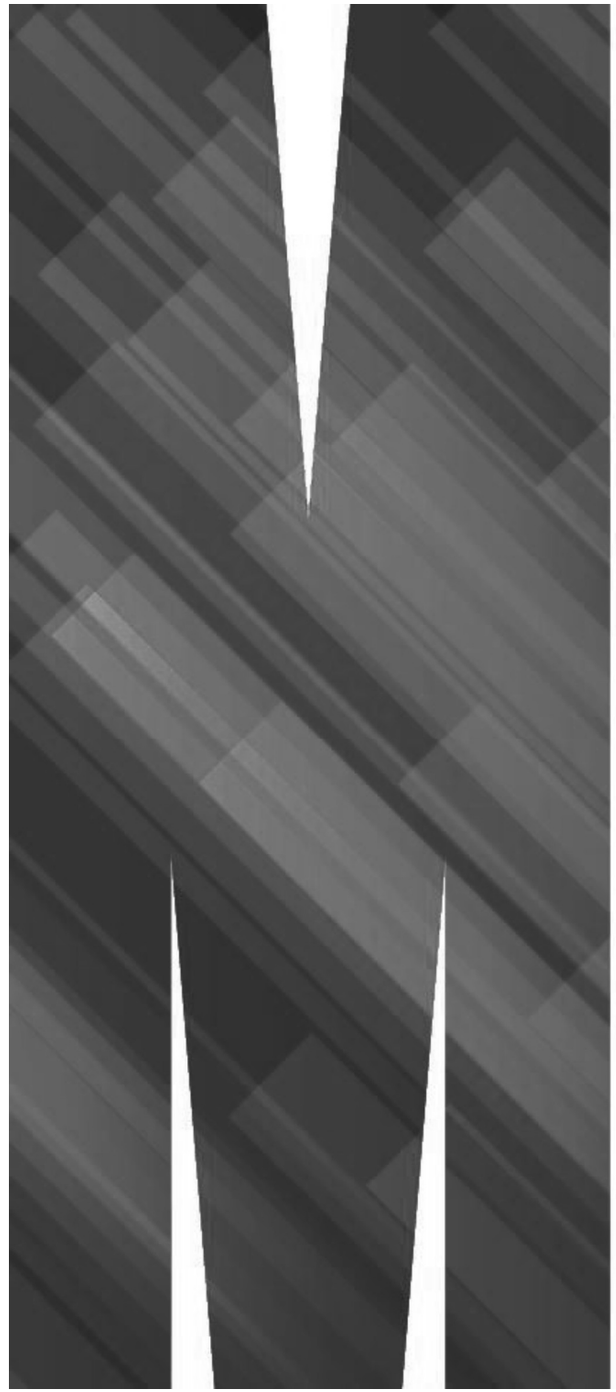
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Topic 6.5
Regression Models

Odds ratios and interpretation of logistic regression models

By Dr Lee How Chinh



Objectives

- Explain odd ratios and use it for interpreting logistic regression models.

Odds Ratios

Probability of an outcome

	Outcome		Total
	Yes	No	
Group A	60	20	80
Group B	90	10	100
Total	150	30	180

Probability of a Yes in Group B

= Total Yes outcomes in Group B ÷ Total outcomes in Group B

= $90 \div 100 = 0.9$

Odds of an outcome

	Outcome		Total
	Yes	No	
Group A	60	20	80
Group B	90	10	100
Total	150	30	180

Odds of a Yes in Group B

= Probability of Yes in Group B ÷ Probability of No in Group B

= $0.9 \div 0.1 = 9$

Odds Ratio of an outcome

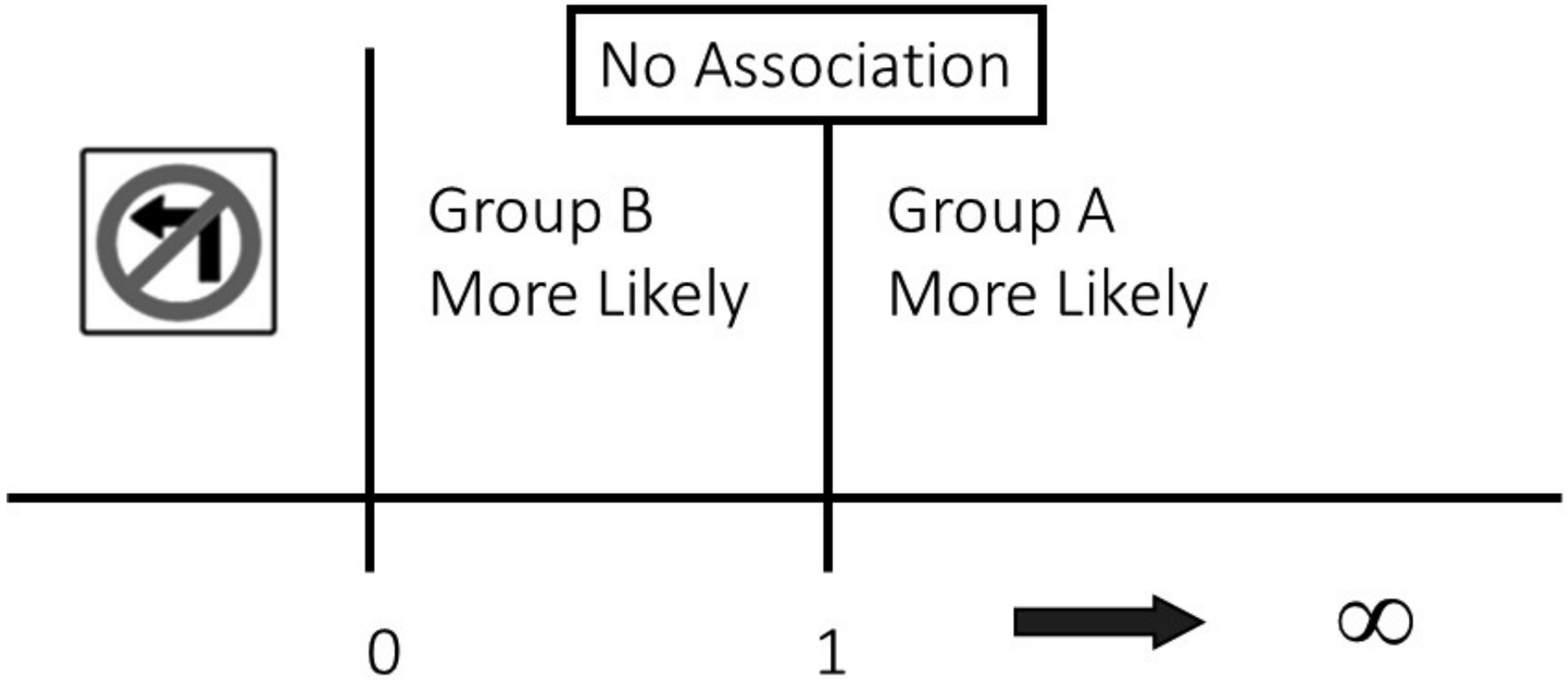
	Outcome		
	Yes	No	Total
Group A	60	20	80
Group B	90	10	100
Total	150	30	180

Odds Ratio, Group A to Group B

= Odds of Yes in Group A ÷ Odds of Yes in Group B

= $3 \div 9 = 0.3333$

Properties of the Odds Ratio, A to B



Interpreting Logistic regression models

Odds Ratio

- An odds ratio expresses the increase in primary outcome odds associated with a unit change in an input.
- It is obtained by taking the exponential of the parameter estimate of the input of interest.

$$\text{logit}(\hat{p}) = \log(\text{odds}) = \beta_0 + \beta_1 x_1$$

$$odds_{\Delta 1} = e^{\beta_0 + \beta_1 (x_1 + 1)}$$

$$odds = e^{\beta_0 + \beta_1 (x_1)}$$

$$\text{Odds Ratio} = \frac{e^{\beta_0 + \beta_1 (x_1 + 1)}}{e^{\beta_0 + \beta_1 (x_1)}} = e^{\beta_1}$$

A Doubling Amount

- A doubling amount gives the amount of change required for doubling the primary outcome odds.
- It is equal to $\log_e(2) \approx 0.69$ divided by the parameter estimate of the input of interest.

$$2odds = e^{\beta_0+\beta_1(x_1+\Delta x_1)}$$
$$odds = e^{\beta_0+\beta_1(x_1)}$$
$$2 = e^{\beta_1(\Delta x_1)}$$

Input need to change to double the odds,

$$\Delta x_1 = \frac{0.69}{\hat{\beta}_1}$$

Interpreting A Logistic Regression Models

For numeric inputs,

IMP_DemAffl

- the odds ratio estimate is 1.283
- This means that for each unit increment of IMP_DemAffl, the odds of the interest response will increase by 28.3%

704	Odds Ratio Estimates		
705			
706			Point
707	Effect		Estimate
708			
709	IMP_DemAffl		1.283
710	IMP_DemAge		0.947
711	IMP_DemGender	F vs U	6.967
712	IMP_DemGender	M vs U	2.899
713	M_DemAffl	0 vs 1	0.708
714	M_DemAge	0 vs 1	0.796
715	M_DemGender	0 vs 1	0.685

Interpreting A Logistic Regression Models

For numeric inputs,

IMP_DemAge

- the odds ratio estimate is 0.947
- This means that for each unit increment of IMP_DemAge, the odds of the interest response will decrease by 5.3%

704	Odds Ratio Estimates		
705			
706			Point
707	Effect		Estimate
708			
709	IMP_DemAff1		1.283
710	IMP_DemAge		0.947
711	IMP_DemGender	F vs U	6.967
712	IMP_DemGender	M vs U	2.899
713	M_DemAff1	0 vs 1	0.708
714	M_DemAge	0 vs 1	0.796
715	M_DemGender	0 vs 1	0.685

Interpreting A Logistic Regression Models

For categorical inputs,

IMP_DemGender (M vs U)

- the odds ratio estimate is 2.899
- This means that for cases with a M value for IMP_DemGender, the odds of interest response are 2.899 times higher than the odds of interest response with a U value for IMP_DemGender

	Odds Ratio Estimates		
	Effect		Point Estimate
	IMP_DemAff1		1.283
	IMP_DemAge		0.947
	IMP_DemGender	F vs U	6.967
	IMP_DemGender	M vs U	2.899
	M_DemAff1	0 vs 1	0.708
	M_DemAge	0 vs 1	0.796
	M_DemGender	0 vs 1	0.685

Lesson Summary

- Odds ratio, A to B > 1, then event A is more likely to happen
- Odds ratio, A to B = 1, then no association between event A and B
- Odds ratio, A to B < 1, then event B is more likely to happen
- Odds ratio of the primary (interest) outcome when an input increases one unit is given by the exponential of the parameter estimate, e^{β_1}.
- The amount of the input need to change to double the odds is equal to 0.69 divided by the parameter estimate, $\Delta x_1 = \frac{0.69}{\widehat{\beta}_1}$

How can I decrease input variables' redundancy in logistic regression models?



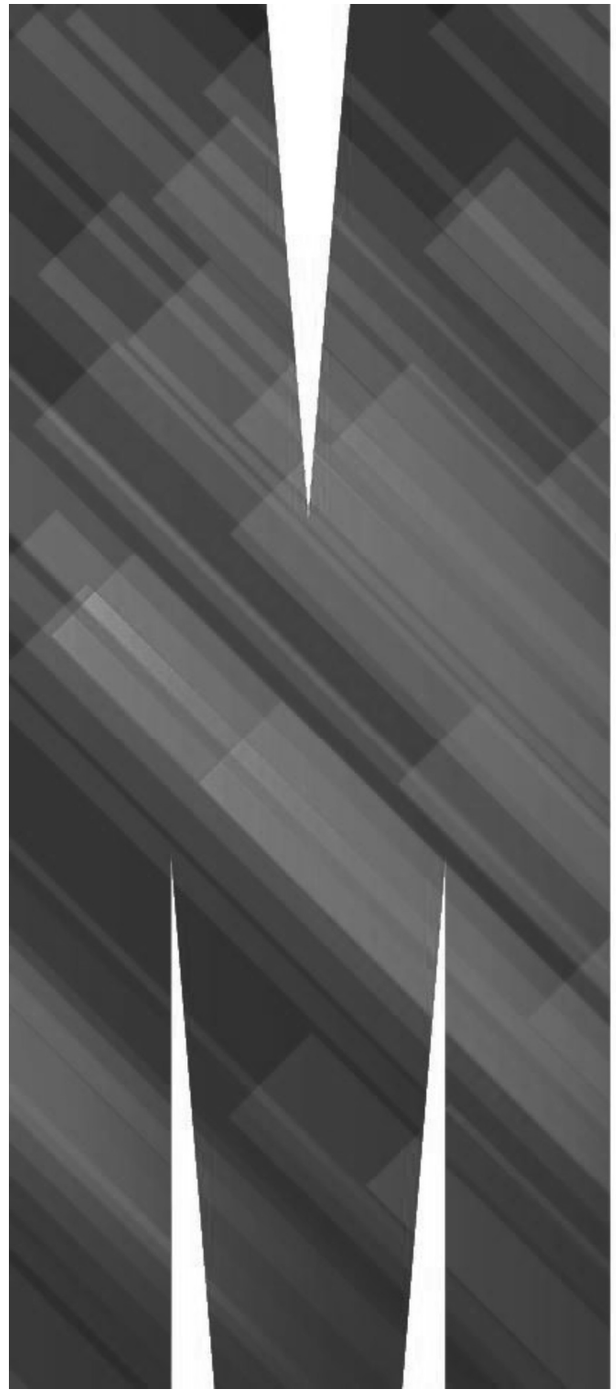
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Topic 6.6
Regression Models

Variable clustering: removing collinearity

By Dr Lee How Chinh



Objectives

- Describe the problem of redundant variables.
- Explain the concept of variable clustering
- Apply variable clustering to remove collinearity

Redundancy

Redundancy

Including redundant inputs in a regression model can cause the following conditions:

- destabilizing the parameter estimates
- increasing the risk of overfitting
- confounding interpretation
- increasing computation time
- increasing scoring effort

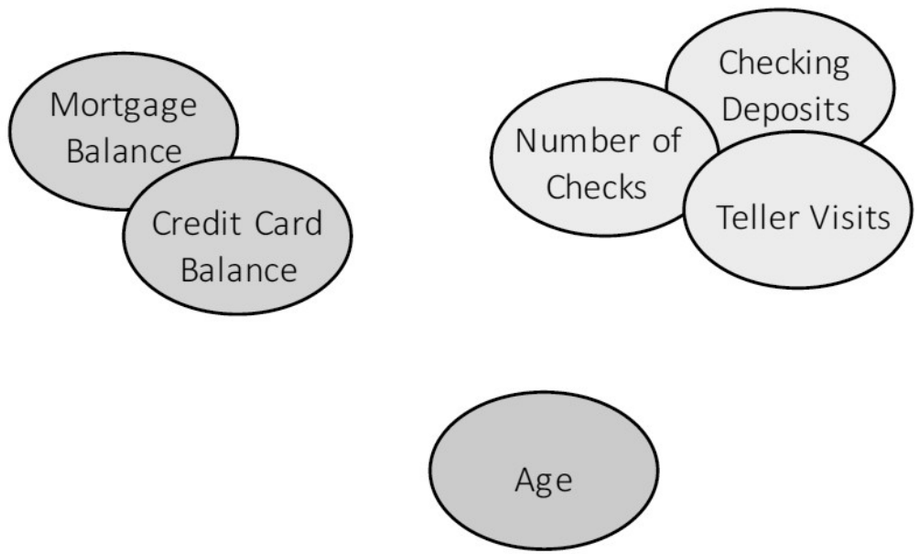
Redundancy

- Redundancy is an unsupervised concept because it does not involve the response (target) variable.
- On the other hand, the relevancy of a variable is the relationship between an input and the response variable.
- In high dimensional data sets, identifying irrelevant inputs is more difficult than identifying redundant inputs.
- A good strategy is to first reduce redundancy and then tackle irrelevancy in a lower dimension space.
- One of the approaches to variable reduction is variable clustering.

Variable Clustering

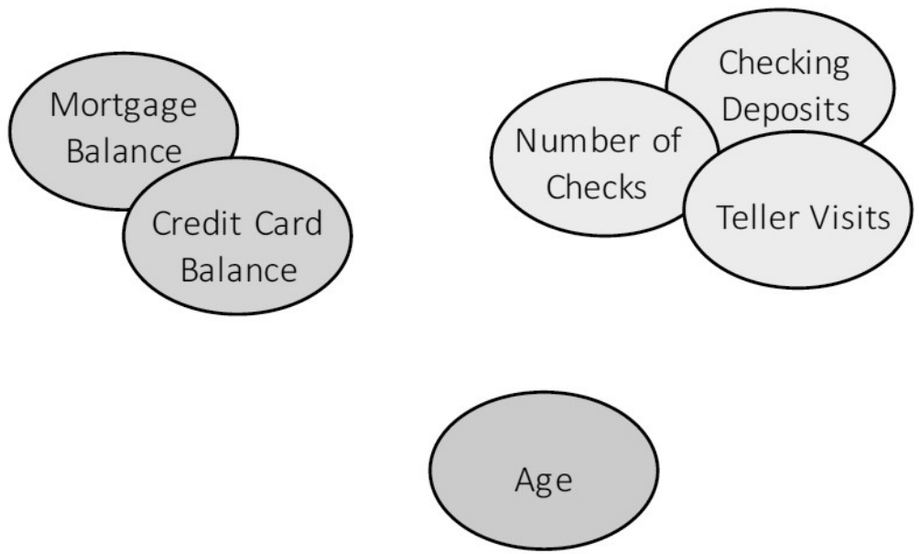
Variable Clustering

- Variable clustering identifies groups of variables that are as correlated as possible among themselves and as uncorrelated as possible with the variables in other clusters.
- A common strategy is to choose one variable from each cluster based on subject-matter knowledge.



Variable Clustering

- Variable clustering identifies groups of variables that are as correlated as possible among themselves and as uncorrelated as possible with the variables in other clusters.
- A common strategy is to choose one variable from each cluster based on subject-matter knowledge.



Lesson Summary

- Input redundancy is an unsupervised issue that does not involve the target.
- An appropriate procedure is to reduce inputs redundancy first and then only handle irrelevant inputs in high dimensional data.
- Variable clustering is to reduce redundancy in inputs.

How can I deal with the issue where certain levels of categorical variable do not have any outputs in target variable?



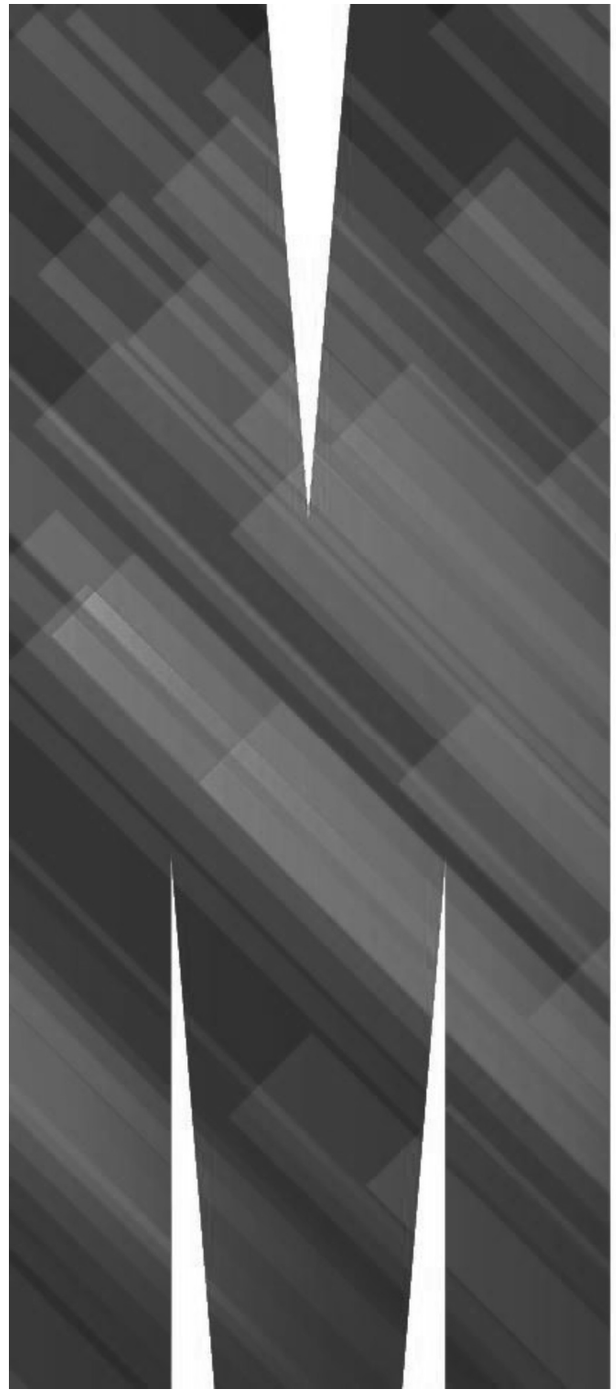
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Topic 6.7
Regression Models

Dealing with Quasi-complete separation

By Dr Lee How Chinh



Objectives

- Identify and deal with quasi-complete separation

Quasi-Complete Separation

Quasi-Complete Separation

- Categorical variables can be included in regression models by introducing dummy variables for the levels.
- However, expanding categorical inputs into dummy variables can increase the dimension of the input space and produce many redundant and irrelevant inputs to the model, and the model may overfit the data.
- Other than that, quasi-complete separation may occur when the target variable is categorical.

Quasi-Complete Separation

- Quasi-complete separation happens when a level of the categorical input has a 0% or 100% of target event.
- One or more of the logits is infinite.
- The likelihood does not have a maximum in at least one dimension, so the maximum likelihood estimate of that coefficient is infinite.

	0	1	D_A	D_B	D_C	logit
A	28	7	1	0	0	-1.39
B	16	0	0	1	0	-Infinity
C	94	11	0	0	1	-2.14
D	23	21	0	0	0	-0.08

- If the zero-cell category is the reference level, then all the coefficients for the dummy variables are infinite.

Quasi-Complete Separation

Quasi-complete separation can lead to

- model interpretation complications
- the estimation algorithm cannot converge.
- misleading in variable selection

The most common cause of quasi-complete separation in predictive modelling with categorical target is categorical inputs with rare categories.

	0	1	D_A	D_B	D_C	logit
A	28	7	1	0	0	-1.39
B	16	0	0	1	0	-Infinity
C	94	11	0	0	1	-2.14
D	23	21	0	0	0	-0.08

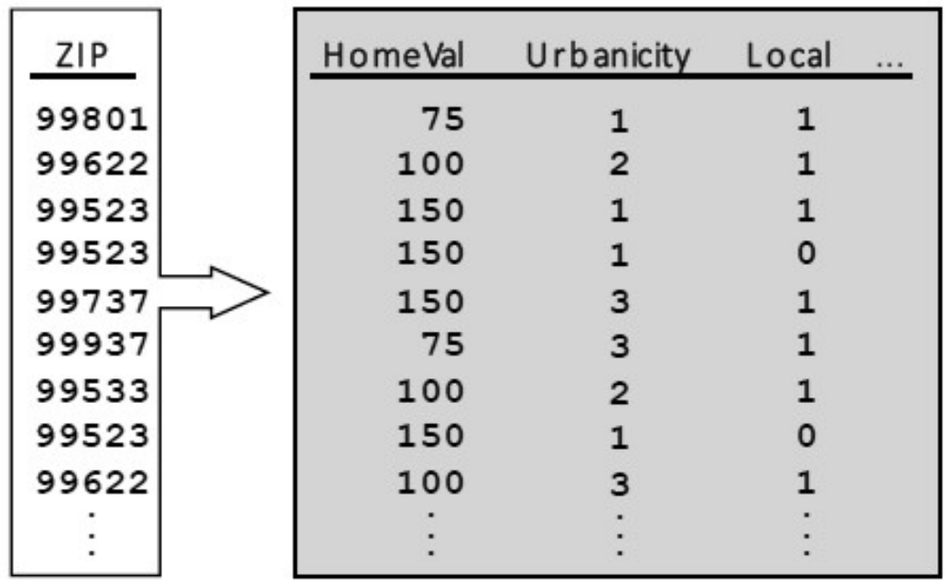
Dealing with Quasi-Complete Separation

Dealing with Quasi-Complete Separation

- Smarter Variable - Use the categorical input as a link to other data sets
- Thresholding - Collapse the categories based on the number of observations in a category.
- Clustering levels - Collapse the categories using some statistical tools.
- Use smoothed weight of evidence coding to convert the categorical input to a continuous input.

Smarter Variables

- Using subject-matter information to create new inputs that represent relevant sources of variation.
- This new categorical input can consider as a link to other inputs.
- For example, geographic areas are often mapped to several relevant demographic variables.



ZIP
99801
99622
99523
99523
99737
99937
99533
99523
99622
⋮

HomeVal	Urbanicity	Local	...
75	1	1	
100	2	1	
150	1	1	
150	1	0	
150	3	1	
75	3	1	
100	2	1	
150	1	0	
100	3	1	
⋮	⋮	⋮	

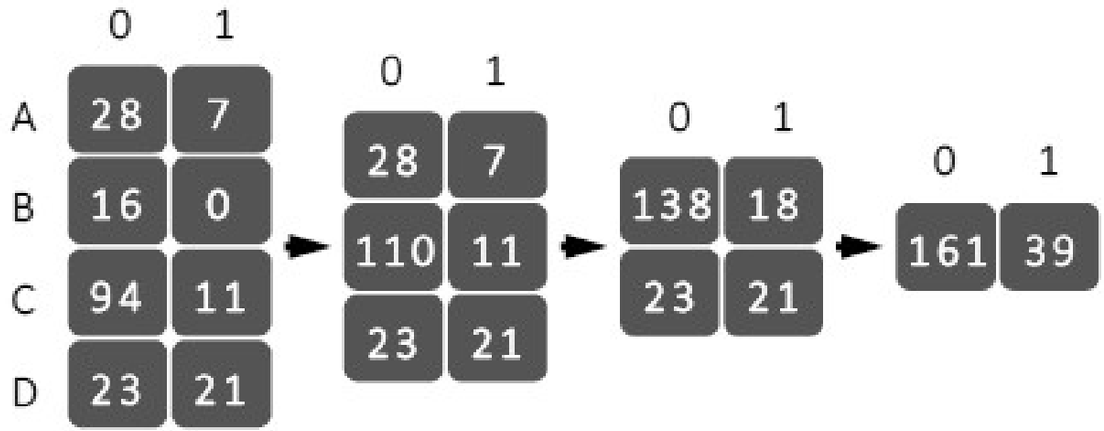
Thresholding

- This method requires a minimum number of cases in a level to create a dummy variable
- For example, the thresholding for the table on the right is 50.
- Any level that fails to meet the minimum threshold is assigned to a new level called OTHER.

Level	Number of observations
A	1562
B	970
C	223
D	111
E	85
F	23
G	17
H	12
I	5

Combine to a single new level "OTHER" and then dummy-code six categories.

Clustering Levels



- Ideally, subject-matter considerations should be used to collapse levels but this is not always practical in predictive modelling.
- Data-driven methods like consolidation tree method can be used to cluster a categorical levels.

Lesson Summary

- Quasi-complete separation occurs when one of the categorical input's levels has a 0% or 100% of the target event.
- Quasi-complete separation can complicate model interpretation, affect the convergence of the estimation and lead to incorrect variable selection.
- Quasi-complete selection can be handled using smarter variables, applying thresholding and clustering categorical levels.

How can I construct a logistic regression model using SAS Viya?



MONASH University